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LEARNER GUIDE

For

Design of Experiment (DOE) Masterclass

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Conducted by

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1.0	2 October 2026	Initial release of the Design of Experiment (DOE) Masterclass learner guide: four topics and ten NovaDOE activities with mock datasets and Excel workbooks.	Dr Alfred Ang

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How to Use This Guide

The slide deck is intentionally concept-led and visual. This guide is the detailed procedural source for the ten NovaDOE activities. Complete each lab in its own folder and retain the exported design, workbook and interpretation.

NovaDOE: https://alfredang.github.io/novadoe/

Course repository: https://github.com/tertiarycourses/C950-Design-of-Experiment-DOE-Masterclass

LMS: https://lms-tms.tertiaryinfotech.com/

Learning Outcomes

- LO1: Analyse process interactions to characterise key factors influencing performance in Design of Experiments.
- LO2: Select suitable factorial DoE projects by evaluating relevant performance metrics.
- LO3: Define the scope and execute fractional factorial DoE projects using problem-solving techniques.
- LO4: Evaluate the effectiveness of DoE projects and recommend follow-up actions.

NovaDOE Workflow

- Select the design method that matches the learning objective.
- Define factors with explicit units, types and feasible bounds.
- Generate the matrix and verify run count, balance and design summary.
- Export CSV and retain both StandardOrder and random RunOrder.
- Join the supplied mock response without changing source values.
- Analyse effects/model fit, interpret limitations and define confirmation.

Topic 01: Fundamentals of Design of Experiments

Questions, variables, variation, hypothesis tests and one-way ANOVA

Key Concepts

- **Engineering experiments:** Learn efficiently from controlled variation. Reduce development time; improve performance; quantify uncertainty; build robust products. *Decision lens: A useful experiment changes a decision, not merely a chart.*
- **DoE lifecycle:** Plan before collecting data. State problem; choose factors and levels; select response; choose design; run; analyse; conclude. *Decision lens: Pre-register the response and decision rule.*
- **Factor, level, treatment, response:** Name the experimental structure. Factor is controlled input; level is setting; treatment is combination; response is measured output. *Decision lens: Units and permissible ranges belong in the factor table.*

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- **Independent and dependent variables:** Separate inputs from outcomes. Independent variables are manipulated; dependent variables respond; nuisance variables are controlled or blocked. *Decision lens: Do not call an observed association an intervention.*
- **OFAT versus DoE:** Interactions require coordinated changes. OFAT is intuitive but inefficient; factorial DoE estimates several effects; interactions are visible; prediction covers the region. *Decision lens: Use OFAT only for safe preliminary range-finding.*
- **Randomisation, replication and blocking:** Protect inference from bias and noise. Randomise run order; replicate treatments; block known nuisance sources; retain run-order evidence. *Decision lens: Repeats measure equipment precision; replicates estimate process variation.*
- **Measurement systems:** Bad measurements cannot be rescued statistically. Define method; verify resolution; control calibration; use consistent units; record missingness. *Decision lens: Measurement error can hide real effects or create false ones.*
- **Hypothesis testing:** Translate evidence into a decision rule. Write null and alternative; choose statistic; calculate p-value; compare with alpha; interpret. *Decision lens: A p-value is conditional on the null and assumptions.*
- **Type I and Type II errors:** Balance false alarms and missed effects. Type I rejects a true null; Type II misses a real effect; power depends on effect, noise and sample size. *Decision lens: Choose risk thresholds from consequence, not habit.*
- **Two-sample comparisons:** Match the test to the sampling design. Independent groups differ from paired measurements; inspect variance; report effect and uncertainty. *Decision lens: Statistical significance does not guarantee practical value.*
- **One-way ANOVA:** Compare more than two treatment means. Partition total variation; estimate between and within variation; compute F ratio; test equal means. *Decision lens: Follow a significant omnibus test with planned comparisons.*
- **Sum-of-squares decomposition:** Account for where variation comes from. SST equals model plus error; degrees of freedom scale mean squares; F compares signal to noise. *Decision lens: Balanced designs make decomposition easier to audit.*

DOE Lab 01: Characterise a Composite Cure Process

Alignment	LO1
NovaDOE method	NovaDOE full factorial planning workspace
Folder	activities/DOE-Lab-01-process-characterisation/
Expected output	A signed experiment charter, factor table, 2^2 design CSV and run-order evidence.
Response	Tensile strength (MPa)

Scenario. A materials team must reduce cure-cycle variation before testing a new composite panel.

Goal. Frame a measurable DoE project and distinguish factors, levels, responses and nuisance variables.

Detailed Procedure

1	**Open NovaDOE and choose Full Factorial.** Use the course NovaDOE URL; do not

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	enter response values yet.
2	**Create the two numeric factors.** Set temperature 160–180 °C and time 45–75 min.
3	**Generate the four-run design and inspect coded levels.** Confirm every low/high combination occurs once.
4	**Export the design as CSV into the lab folder.** Name it learner-design.csv.
5	**Complete the charter sheet in the workbook.** State problem, response, unit, objective, constraints, nuisance variables and decision rule.
6	**Assign a random run order and record planned evidence.** Do not overwrite StandardOrder.
7	**Review the design with a peer before execution.** Use the scope, safety and measurement checklist.

Acceptance Test

The factor table, four treatment combinations, response definition, random run order and go/no-go rule are all present.

Evidence to Retain

- NovaDOE setup screenshot and learner-design.csv export.
- Completed DOE-Lab-Workbook.xlsx with formulas and interpretation.
- A concise recommendation, limitation and follow-up action.

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DOE Lab 02: Compare Coating Cure Temperatures with One-Way ANOVA

Alignment	LO1
NovaDOE method	NovaDOE full factorial (one 4-level categorical factor) + Excel ANOVA
Folder	activities/DOE-Lab-02-one-way-anova/
Expected output	A randomised 4-level design, replicated response dataset, ANOVA table and engineering interpretation.
Response	Adhesion strength (N/mm)

Scenario. A coating line compares four cure temperatures using adhesion strength from replicated coupons.

Goal. Separate between-treatment and within-treatment variation and interpret an ANOVA decision.

Detailed Procedure

1	**Choose Full Factorial in NovaDOE.** Create a categorical factor with 140, 150, 160 and 170 °C levels.
2	**Generate the design and export CSV.** Retain the level labels.
3	**Open mock-data.csv and inspect the five replicates per level.** Flag any missing or impossible values.
4	**Paste or link the data into the workbook Raw_Data sheet.** Keep the source table unchanged.
5	**Calculate group count, mean and standard deviation.** Use Excel formulas in the Analysis sheet.
6	**Run Data Analysis > ANOVA: Single Factor.** Use alpha = 0.05 and include labels.
7	**Interpret F, p-value and practical difference.** State whether follow-up comparisons are warranted.

Acceptance Test

The ANOVA uses 20 observations, reports the correct factor levels and gives a conclusion that separates statistical from practical significance.

Evidence to Retain

- NovaDOE setup screenshot and learner-design.csv export.
- Completed DOE-Lab-Workbook.xlsx with formulas and interpretation.
- A concise recommendation, limitation and follow-up action.

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Topic 02: Factorial Design of Experiments

Main effects, interactions, regression, ANOVA, blocks and confounding

Key Concepts

- **Factorial design:** Test combinations rather than isolated changes. All combinations estimate main effects; interaction estimates become possible; one dataset supports prediction. *Decision lens: Keep the experimental region safe and relevant.*
- **The 2^2 design:** Two factors at low and high levels. Four treatment combinations; coded levels -1 and +1; geometric square; balanced contrasts. *Decision lens: Add replicates or centre points when pure error or curvature matters.*
- **Main effects:** Average response change across another factor. Compare high versus low means; sign gives direction; magnitude gives practical influence; uncertainty qualifies the estimate. *Decision lens: Averaging can conceal a strong interaction.*
- **Interactions:** The effect of one factor depends on another. Non-parallel interaction lines; product term in regression; hierarchy principle; joint settings drive the decision. *Decision lens: Never interpret main effects alone when interaction is material.*
- **Coded regression model:** Connect design geometry to prediction. Intercept is grand mean; coefficients represent half-effects; product term models interaction; residuals test adequacy. *Decision lens: Decode recommendations back to engineering units.*
- **ANOVA for factorial designs:** Test model terms against residual noise. Degrees of freedom follow effects; mean square divides SS by df; F and p assess evidence; lack-of-fit needs replication. *Decision lens: Report effect size and confidence intervals alongside p-values.*
- **The 2^3 cube:** Three factors reveal two- and three-way interactions. Eight corners; three main effects; three two-way interactions; one three-way interaction. *Decision lens: A high-order interaction is hard to explain and needs engineering evidence.*
- **General 2^k designs:** Run count doubles with every factor. 2^k treatment combinations; orthogonal effect estimates; effect hierarchy guides model reduction; cost grows exponentially. *Decision lens: Use fractional designs when screening many factors.*
- **Pareto and normal effect plots:** Prioritise candidate effects. Standardised effects show magnitude; reference line guides screening; normal plot separates signal from noise. *Decision lens: Plots support, but do not replace, model assumptions.*
- **Centre points and curvature:** Check whether a straight plane is adequate. Centre runs estimate local curvature; replicated centres provide pure error; significant curvature motivates RSM. *Decision lens: A centre point does not identify which quadratic term matters.*
- **Replication strategy:** Create an honest error term. Independent replicates capture process variation; duplicate measurements capture measurement variation; power increases with information. *Decision lens: Do not treat repeated readings from one specimen as independent runs.*
- **Blocking:** Remove known nuisance variation. Group homogeneous runs; randomise within blocks; estimate block effect; preserve treatment balance. *Decision lens: A block should not be confused with a factor of scientific interest.*

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- **Confounding in blocks:** Trade an interaction for practical run groups. Choose defining contrast; aliases follow; high-order terms are typical sacrifices; document the alias structure. *Decision lens: Never confound an effect that stakeholders consider critical.*

DOE Lab 03: Reduce Order-Processing Time with a 2^2 Factorial

Alignment	LO2
NovaDOE method	NovaDOE full factorial
Folder	activities/DOE-Lab-03-full-factorial-2x2/
Expected output	A 12-run replicated design, effect table, interaction interpretation and recommended setting.
Response	Processing time (hours)

Scenario. A distribution centre tests Current/New ordering systems and Packing A/B to minimise processing hours.

Goal. Estimate two main effects and their interaction from a replicated 2^2 experiment.

Detailed Procedure

1	**Select Full Factorial in NovaDOE.** Add two categorical factors with two levels each.
2	**Set three replicates and generate the design.** If the UI has no replicate field, export once and duplicate treatment combinations in Excel.
3	**Export CSV and join the provided Hours values by standard order.** Never join by the random run number.
4	**Calculate treatment means and high-minus-low effects.** Use formulas and show the sign convention.
5	**Create a main-effects chart.** Keep the vertical scale identical across panels.
6	**Create an interaction chart.** Compare slopes and explain non-parallel lines.
7	**Fit a coded model including A, B and A×B.** Retain hierarchy if the interaction is kept.
8	**Recommend the lowest-time combination with evidence.** Include a confirmation-run plan.

Acceptance Test

All four combinations have three independent replicates and the recommendation cites both the predicted response and interaction risk.

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Evidence to Retain

- NovaDOE setup screenshot and learner-design.csv export.
- Completed DOE-Lab-Workbook.xlsx with formulas and interpretation.
- A concise recommendation, limitation and follow-up action.

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DOE Lab 04: Model a 2^3 Composite Bonding Experiment

Alignment	LO2
NovaDOE method	NovaDOE full factorial
Folder	activities/DOE-Lab-04-full-factorial-2k/
Expected output	An eight-corner cube design, effect hierarchy, coded regression and predicted optimum.
Response	Bond strength (MPa)

Scenario. A bonding team studies temperature, pressure and surface preparation to maximise joint strength.

Goal. Analyse main and interaction effects in a three-factor design and select a practical model.

Detailed Procedure

1	**Choose Full Factorial and add three factors.** Use two numeric and one categorical factor.
2	**Generate the eight-run design and export CSV.** Verify $2^3 = 8$ unique treatment combinations.
3	**Join the response values from mock-data.csv.** Preserve RunOrder and StandardOrder.
4	**Code numeric and categorical low/high levels as -1/+1.** Document the coding table.
5	**Calculate A, B, C, AB, AC, BC and ABC contrast estimates.** Use auditable worksheet formulas.
6	**Build a Pareto-style effect chart.** Label effect magnitudes and direction separately.
7	**Fit a hierarchical coded regression model.** Inspect residuals before reducing terms.
8	**Predict and confirm the proposed operating combination.** State limits of the two-level region.

Acceptance Test

The workbook contains all seven effect estimates, a hierarchical model, residual check and a recommendation in engineering units.

Evidence to Retain

- NovaDOE setup screenshot and learner-design.csv export.
- Completed DOE-Lab-Workbook.xlsx with formulas and interpretation.
- A concise recommendation, limitation and follow-up action.

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DOE Lab 05: Block a Replicated Seal-Strength Experiment

Alignment	LO2
NovaDOE method	NovaDOE full factorial + blocked run sheet
Folder	activities/DOE-Lab-05-blocking-replication/
Expected output	Two balanced blocks, randomisation within block, block-adjusted effects and deviation log.
Response	Seal strength (N)

Scenario. Seal trials span morning and afternoon shifts whose ambient conditions differ.

Goal. Use blocking to remove shift-to-shift variation without sacrificing treatment balance.

Detailed Procedure

1	**Generate a 2^2 full factorial in NovaDOE.** Export the four standard treatments.
2	**Duplicate the design into two blocks.** Name blocks AM and PM; do not change factor combinations.
3	**Randomise the four runs independently inside each block.** Use a fixed recorded seed or Excel RAND with values frozen.
4	**Import the response dataset.** Check every block has each treatment once.
5	**Estimate treatment and block means.** Use formulas with visible denominators.
6	**Compare models with and without the block term.** Explain what variation the block absorbs.
7	**Review the deviation log.** Decide whether any run needs investigation, not silent deletion.

Acceptance Test

Each block contains all four treatments, run order is randomised within block and the block-adjusted conclusion is documented.

Evidence to Retain

- NovaDOE setup screenshot and learner-design.csv export.
- Completed DOE-Lab-Workbook.xlsx with formulas and interpretation.
- A concise recommendation, limitation and follow-up action.

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Topic 03: Fractional Factorial and Screening Designs

Generators, aliases, resolution, Plackett-Burman and sequential experimentation

Key Concepts

- **Why fractionate:** Screen many factors with fewer runs. Assume effect sparsity; select a defensible fraction; preserve critical estimates; reserve budget for confirmation. *Decision lens: A fraction is a strategic information trade, not free efficiency.*
- **One-half fractions:** Use a generator to select half the full design. Generator defines included runs; defining relation creates aliases; signs determine fraction; replicate if error is needed. *Decision lens: Record the generator so the design is reproducible.*
- **Alias structure:** Know which effects are inseparable. Multiply effects by defining words; aliases share estimates; resolution summarises shortest word; subject knowledge breaks ambiguity. *Decision lens: Do not label an aliased estimate as one unique physical cause.*
- **Resolution III:** Main effects alias with two-factor interactions. Useful for early screening; smallest run count; strong interaction risk; confirmation essential. *Decision lens: Accept only when interactions are plausibly negligible.*
- **Resolution IV:** Main effects clear of two-factor interactions. Main effects alias with three-factor interactions; two-factor interactions alias in pairs; common screening compromise. *Decision lens: Escalate ambiguous interaction pairs in a follow-up design.*
- **Resolution V:** Main effects and two-factor interactions are separately estimable. Main effects alias with four-factor terms; two-factor interactions alias with three-factor terms; more runs. *Decision lens: Choose when interactions matter to the decision.*
- **One-quarter fractions:** Two generators select one fourth of runs. Two defining words plus their product; alias chains lengthen; resolution depends on generator choice; efficiency increases. *Decision lens: Generator selection is part of design quality.*
- **Foldover designs:** Break targeted aliases in a follow-up. Reverse selected factor signs; combine original and foldover; separate main effects from interactions; retain sequential learning. *Decision lens: Plan foldover before consuming the entire budget.*
- **Plackett-Burman screening:** Estimate many main effects economically. N runs screen up to N-1 factors; orthogonal main-effect columns; interactions are not independently estimable; good for factor reduction. *Decision lens: Use results to design a factorial confirmation, not to optimise directly.*
- **Screening metrics:** Select projects and factors using business evidence. Effect magnitude; signal-to-noise; cost per run; cycle time; safety and specification impact. *Decision lens: A statistically active factor may still be operationally irrelevant.*
- **Sequential experimentation:** Learn, narrow and augment. Screen; confirm; map curvature; optimise; verify at the recommended settings. *Decision lens: Every phase needs a go/no-go criterion.*
- **Design review and run sheet:** Make the experiment executable. Scope; owners; factor table; randomised run order; measurement plan; risks; stop rules. *Decision lens: Freeze the run sheet before execution and log deviations.*

DOE Lab 06: Screen Five Factors with a Resolution V Half Fraction

Alignment	LO3
NovaDOE method	NovaDOE fractional factorial

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Folder	activities/DOE-Lab-06-fractional-factorial/
Expected output	A 16-run fraction, generator/alias record, screened effects and confirmation plan.
Response	Defect rate (%)

Scenario. A polymer extrusion team has five controllable factors but only 16 safe pilot runs.

Goal. Generate a defensible 2^(5-1) design, interpret aliases and plan confirmation.

Detailed Procedure

1	**Select Fractional Factorial in NovaDOE.** Add five numeric factors with safe low/high limits.
2	**Choose a 16-run fraction and record the displayed resolution.** Target Resolution V where supported.
3	**Generate and export the design CSV.** Save the NovaDOE setup in browser storage as well.
4	**Import the mock response and code factor levels.** Lower defect rate is better.
5	**Calculate main-effect contrasts.** Rank absolute effects and retain signs.
6	**Write the generator and alias interpretation in the workbook.** If NovaDOE does not display aliases, document the assumed generator from the exported matrix.
7	**Select factors for a confirmation experiment.** Use magnitude, feasibility and engineering plausibility.
8	**Plan confirmation and possible foldover runs.** Define expected improvement and acceptance threshold.

Acceptance Test

The design has 16 unique runs, five factor columns, a stated resolution/alias assumption and a confirmation plan that does not overclaim aliased effects.

Evidence to Retain

- NovaDOE setup screenshot and learner-design.csv export.
- Completed DOE-Lab-Workbook.xlsx with formulas and interpretation.
- A concise recommendation, limitation and follow-up action.

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DOE Lab 07: Screen Eleven Battery-Process Factors

Alignment	LO3
NovaDOE method	NovaDOE Plackett-Burman
Folder	activities/DOE-Lab-07-plackett-burman/
Expected output	A 12-run PB matrix, main-effect ranking, risk register and factorial follow-up proposal.
Response	Capacity retention (%)

Scenario. A battery lab must screen eleven formulation and process factors in twelve runs.

Goal. Use a Plackett-Burman design to reduce a long factor list before modelling interactions.

Detailed Procedure

1	**Select Plackett-Burman in NovaDOE.** Create eleven two-level numeric factors.
2	**Generate a 12-run design.** Confirm the summary reports 12 runs and 11 factors.
3	**Export the design and join mock capacity-retention results.** Keep factors in the same order.
4	**Calculate each main-effect contrast.** Use the high-level average minus low-level average.
5	**Create a ranked effect chart.** Show absolute magnitude and sign.
6	**Assess interaction contamination risk.** Use process knowledge to identify plausible two-factor interactions.
7	**Choose three or four factors for follow-up.** Balance evidence, controllability, safety and cost.
8	**Draft a full or fractional factorial confirmation.** Do not optimise from the PB model alone.

Acceptance Test

All eleven main effects are ranked and the report explicitly states that interactions are not separately estimable in the screening design.

Evidence to Retain

- NovaDOE setup screenshot and learner-design.csv export.
- Completed DOE-Lab-Workbook.xlsx with formulas and interpretation.
- A concise recommendation, limitation and follow-up action.

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Topic 04: Screening, Modelling and Optimising DoE

Response surfaces, CCD, Box-Behnken, mixtures and Taguchi robustness

Key Concepts

- **Response Surface Methodology:** Model curvature after important factors are known. Fit first- and second-order terms; map a local region; find stationary behaviour; optimise with constraints. *Decision lens: RSM is local—avoid extrapolation beyond the design space.*
- **Quadratic models:** Curvature needs squared terms. Linear terms give slope; interaction terms twist the surface; squared terms create curvature; residuals test adequacy. *Decision lens: Respect hierarchy when retaining higher-order terms.*
- **Central Composite Design:** Augment factorial points with stars and centres. Factorial corners estimate interactions; axial points estimate curvature; centre points estimate pure error; alpha controls geometry. *Decision lens: CCD supports efficient sequential augmentation.*
- **CCD variants and alpha:** Choose geometry for feasibility and prediction. Circumscribed extends beyond factorial range; inscribed keeps axial points within bounds; face-centred uses alpha 1; rotatability stabilises variance. *Decision lens: Safety limits may rule out circumscribed designs.*
- **Box-Behnken design:** Estimate quadratic models without all-factor extremes. Mid-edge treatment combinations; centre points; at least three factors; fewer dangerous corner runs. *Decision lens: Compare run count and feasible region with CCD.*
- **Contour and surface plots:** Translate equations into operating windows. Contours show equal-response lines; surfaces show curvature; overlay multiple requirements; mark the data region. *Decision lens: A persuasive plot still needs uncertainty and validation.*
- **Optimisation and confirmation:** Turn a model into a controlled recommendation. Define target and bounds; optimise desirability; inspect sensitivity; run confirmation trials; compare predicted and observed. *Decision lens: Do not release settings without confirmation.*
- **Mixture designs:** Component proportions sum to one. Changing one component forces others to change; simplex geometry replaces a cube; blends drive the response; process variables can be added. *Decision lens: Ordinary factorial independence does not apply to proportions.*
- **Simplex centroid and extreme vertices:** Cover constrained formulation space. Vertices are pure blends; edges are binary blends; centroid is balanced blend; constraints reshape the feasible region. *Decision lens: Document lower and upper component bounds.*
- **Taguchi robust design:** Reduce sensitivity to uncontrollable noise. Separate control and noise factors; use orthogonal arrays; calculate signal-to-noise ratio; tune mean after stabilising variation. *Decision lens: Robustness is performance across noise, not merely a high mean.*
- **Signal-to-noise ratios:** Match the metric to the engineering goal. Smaller-the-better; larger-the-better; nominal-the-best; logarithmic ratio combines mean and variation. *Decision lens: State the S/N convention before comparing settings.*

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- **Evaluate and recommend:** Close the experimental learning loop. Compare objective to achieved effect; check uncertainty and residuals; document limitations; assign actions; monitor control metrics. *Decision lens: Recommendation strength must match evidence strength.*
- **Follow-up action portfolio:** Choose the next experiment intentionally. Confirm promising settings; fold over aliases; augment for curvature; widen or narrow ranges; improve measurement system. *Decision lens: Capture decision, owner, due date and acceptance test.*

DOE Lab 08: Design a Robust Golf-Ball Process with Taguchi Logic

Alignment	LO4
NovaDOE method	NovaDOE full/fractional matrix used as an orthogonal-array proxy
Folder	activities/DOE-Lab-08-taguchi-robustness/
Expected output	An orthogonal eight-run control matrix, two noise responses, S/N analysis and robustness recommendation.
Response	Flight distance (m) under two clubs

Scenario. A golf-equipment team tests four two-level control factors under Driver and 5-Iron noise conditions.

Goal. Separate control and noise factors and recommend robust rather than merely high-mean settings.

Detailed Procedure

1	**Generate an eight-run balanced two-level matrix in NovaDOE.** Use Fractional Factorial for four control factors.
2	**Export CSV and verify level balance.** Each level should occur equally often per factor.
3	**Join Driver and 5-Iron distance responses.** Treat club as a planned noise condition.
4	**Calculate run mean and standard deviation.** Use formulas and consistent units.
5	**Calculate larger-the-better S/N ratio.** Use -10*LOG10(AVERAGE(1/y²)).
6	**Create factor-response tables for mean and S/N.** Rank delta across levels.
7	**Choose settings that maximise S/N, then tune mean.** Explain any trade-off.
8	**Propose a confirmation trial across both noise conditions.** Set robustness acceptance criteria.

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Acceptance Test

The recommended settings are justified by S/N ratio and mean response, with a confirmation plan across both club conditions.

Evidence to Retain

- NovaDOE setup screenshot and learner-design.csv export.
- Completed DOE-Lab-Workbook.xlsx with formulas and interpretation.
- A concise recommendation, limitation and follow-up action.

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DOE Lab 09: Optimise Sterile-Package Seal Strength with CCD

Alignment	LO4
NovaDOE method	NovaDOE Central Composite
Folder	activities/DOE-Lab-09-central-composite/
Expected output	A CCD export, quadratic model table, contour evidence, optimum and confirmation record.
Response	Seal strength (N)

Scenario. A medical-packaging engineer targets seal strength of 26 N within an acceptable 20–32 N range.

Goal. Generate a CCD, fit a quadratic response surface and identify a feasible operating window.

Detailed Procedure

1	**Select Central Composite in NovaDOE.** Add three numeric factors with low/high bounds.
2	**Choose face-centred or circumscribed geometry based on safe limits.** Record the chosen CCD type and centre points.
3	**Generate and export the design.** Verify factorial, axial and centre runs are present.
4	**Join mock seal-strength responses.** Flag values outside 20–32 N without deleting them.
5	**Create linear, interaction and squared columns.** Use coded factor levels.
6	**Fit the quadratic model with Excel Regression or LINEST.** Retain hierarchy and inspect residuals.
7	**Create contour-ready prediction grids.** Keep predictions within the experimental region.
8	**Select settings closest to 26 N with low sensitivity.** Balance target achievement and robustness.
9	**Define and evaluate confirmation trials.** Compare observed with predicted interval.

Acceptance Test

The design includes factorial, axial and centre points; the recommendation targets 26 N and includes confirmation evidence.

Evidence to Retain

- NovaDOE setup screenshot and learner-design.csv export.

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- Completed DOE-Lab-Workbook.xlsx with formulas and interpretation.
- A concise recommendation, limitation and follow-up action.

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DOE Lab 10: Optimise a Three-Component Thermal Paste Mixture

Alignment	LO4
NovaDOE method	NovaDOE Latin Hypercube for candidate exploration + simplex filtering
Folder	activities/DOE-Lab-10-mixture-followup/
Expected output	A constrained candidate set, simplex plot table, blend model evidence and validation plan.
Response	Thermal conductivity (W/m·K)

Scenario. A thermal-materials lab blends ceramic filler, binder and additive; proportions must total 100%.

Goal. Recognise the mixture constraint, map a simplex design and recommend a constrained follow-up.

Detailed Procedure

1	**Select Latin Hypercube in NovaDOE.** Generate at least 40 candidate points for three numeric proportions.
2	**Export the candidate design.** Treat the values as proportions, not independent production settings.
3	**Normalise each row so the three components sum to 1.** Use formulas and show the original values.
4	**Filter candidates against component bounds.** Retain the feasibility flag.
5	**Select a balanced subset spanning vertices, edges and centre-like blends.** Document the selection rule.
6	**Join the mock response data for the selected blends.** Do not extrapolate to discarded infeasible candidates.
7	**Fit a simple mixture model and compare predictions.** Avoid an intercept if using canonical mixture terms.
8	**Recommend a blend and follow-up extreme-vertices experiment.** State composition tolerance and validation tests.
9	**Review the full portfolio of ten labs.** Link the outcome to an evaluation of effectiveness and follow-up actions.

Acceptance Test

Every retained blend sums to 1.000 within tolerance, satisfies bounds and leads to a constrained follow-up recommendation.

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Evidence to Retain

- NovaDOE setup screenshot and learner-design.csv export.
- Completed DOE-Lab-Workbook.xlsx with formulas and interpretation.
- A concise recommendation, limitation and follow-up action.

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Troubleshooting

- **NovaDOE design differs:** Check design method, factor type, level order, run count and saved setup.
- **CSV joins incorrectly:** Join by StandardOrder and keep RunOrder as execution evidence.
- **Effect direction reverses:** Verify high-minus-low coding before interpreting the sign.
- **Model overfits:** Use hierarchy, residual evidence and engineering plausibility.
- **Mixture total is not 1:** Normalise proportions, then re-check all component bounds.

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